

The Foundations: Logic and Proofs

Chapter 1, Part I: Propositional Logic

Propositional Logic

Section 1.1

1.1.2 Propositions (命题)

A *proposition* is a declarative sentence that is either true or false, but not both.

Examples of propositions:

- | | |
|--------------------------------------|---|
| a) Toronto is the capital of Canada. | F |
| b) $1 + 0 = 1$ | T |
| c) $0 + 0 = 2$ | F |

Examples that are not propositions.

- a) Sit down!
- b) What time is it?
- c) $x + 1 = 2$
- d) $x + y = z$

Declarative 陈述式

Propositional Logic

Constructing Propositions

- **Propositional Variables**(命题变数): $p, q, r, s, \dots$
- The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.
- **Compound Propositions**(复合命题); constructed from **logical connectives**(逻辑连词) and **other propositions**
 - Negation $\neg$ (非)
 - Conjunction $\wedge$ (与)
 - Disjunction $\vee$ (或)
 - Implication $\rightarrow$ (蕴含)
 - Biconditional $\leftrightarrow$ (双条件)

Compound Propositions: Negation(非)

The *negation* of a proposition p is denoted by $\neg p$ (“not p ”). The truth value of $\neg p$ is the opposite of the truth value of p . It has this truth table (真值表):

p	$\neg p$
T	F
F	T

Example: If p denotes “The earth is round.”, then $\neg p$ denotes “It is not the case that the earth is round,” or more simply “The earth is not round.”

Conjunction (合取式, 与)

The *conjunction* of propositions p and q is denoted by $p \wedge q$ (p and q). $p \wedge q$ is **true when both p and q are true and is false otherwise** and has this truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

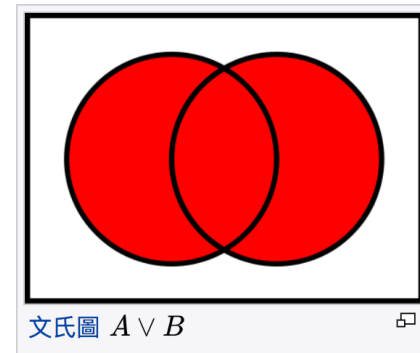
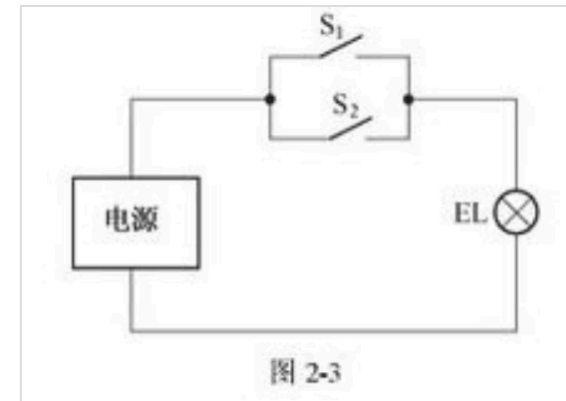
Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home and it is raining.”

Disjunction (分取式, 或)

The *disjunction* of propositions p and q is denoted by $p \vee q$ (p or q). $p \vee q$ is false when both p and q are false and is true otherwise and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes “I am at home or it is raining.”



The Connective Or in English

In English “or” has two distinct meanings.

- “**Inclusive Or**”(包含或) - In the sentence “Students who have taken MATH101 or MA001 may take this class,” we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of **disjunction**. For $p \vee q$ to be true, either one or both of p and q must be true.
- “**Exclusive Or**”(异或) - When reading the sentence “Soup or salad comes with this entrée,” we do not expect to be able to get both soup and salad. This is the meaning of **Exclusive Or (Xor)**. In $p \oplus q$ ($p \text{ XOR } q$), one of p and q must be true, but not both. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

1.1.3 Implication (蕴含)

If p and q are propositions, then $p \rightarrow q$ is a *conditional statement* (条件语句) or *implication* which is read as “if p , then q ”. $p \rightarrow q$ is false when p is true and q is false, and true otherwise. It has this truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “If I am at home then it is raining.”

In $p \rightarrow q$, p is the *hypothesis* (antecedent or premise) (假设 前提) and q is the *conclusion* (or consequence) (结论).

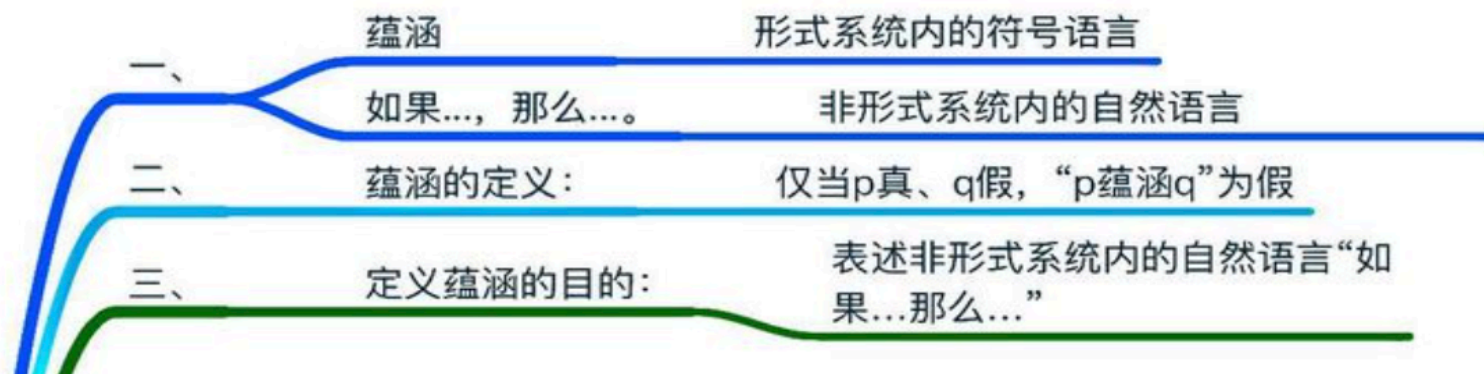
Understanding Implication₁

One way to view the logical conditional is to think of an obligation or contract.

- “If I am elected, then I will lower taxes.”

If the politician is elected, voters would expect this politician to lower taxes. Furthermore, if the politician is not elected, then voters will not have any expectation. If the politician is elected and does not lower taxes, then the voters can say that he or she has broken the campaign pledge. This corresponds to the case where p is true and q is false.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T



1. 在非形式系统中, 我们说“如果p,那么q”为真时, 根据经验和直观判断:

(1) 前提: “如果p,那么q”, p真

结论: q真

(2) 前提: “如果p,那么q”, p假

结论: q的真假不确定

(3) 前提: “如果p,那么q”, q假

结论: p假

(4) 前提: “如果p,那么q”, q真

结论: p的真假不确定

2. 而在形式系统中“p蕴涵q为真”，根据判断系统内有效性的语形有效和语义有效两方面：

(1) 前提：“p蕴涵q”，p真

(4) 前提：“p蕴涵q”，q真

结论：q真

结论：p的真假不确定

(2) 前提：“p蕴涵q”，p假

(4)

结论：q的真假不确定

p	q	p 蕴涵 q	
真	真	真	语法有效
真	假	假	
假	真	真	语法有效
假	假	真	

(3) 前提：“p蕴涵q”，q假

结论：p假

形式系统中前提是“p蕴涵q”，非形式系统中前提是“如果...，那么...”，此时形式系统的有效性与非形式系统的有效性一致，形式系统的“蕴涵”能表述非形式系统的“如果...那么...”。

形式系统中前提是“ p 蕴涵 q ”，非形式系统中前提是“如果...，那么...”，此时形式系统的有效性与非形式系统的有效性一致，形式系统的“蕴涵”能表述非形式系统的“如果...，那么...”。

Different Ways of Expressing $p \rightarrow q$

if p , then q

if p , q

q if p

q whenever p

q follows from p

q unless $\neg p$

a necessary condition for p is q

a sufficient condition for q is p

p is sufficient for q (充分)

q is necessary for p (必要)

p implies q

p only if q (仅当)

q when p (当)

Converse, Contrapositive, and Inverse

From $p \rightarrow q$ we can form new conditional statements .

- $q \rightarrow p$ is the **converse**(逆) of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive**(逆否) of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse**(否) of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It raining is a sufficient condition for my not going to town.”

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

Biconditional (bi-implications)(双条件)

If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise. The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”

Expressing the Biconditional

Some alternative ways “ p if and only if q ” is expressed in English:

- p is **necessary and sufficient** for q (必要充分)
- **if p then q , and conversely**
- p **iff** q

1.1.4 Truth Tables For Compound Propositions

Construction of a truth table:

Rows

- Need a row for every possible combination of values for the atomic propositions简单命题.

Columns

- Need a column for the compound proposition (usually at far right)
- Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
 - This includes the atomic propositions

Example Truth Table

Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

Two propositions are *equivalent* if they always have the same truth value.

Example: Show using a truth table that the conditional statement is equivalent to the contrapositive.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the inverse 否 nor converse 逆 of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Problem

How many rows are there in a truth table with n propositional variables?

Solution: 2^n We will see how to do this in Chapter 6.

Note that this means that with n propositional variables, we can construct 2^n distinct (that is, not equivalent) propositions.

1.1.5 Precedence of Logical Operators (optional)

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow (\neg r)$

If the intended meaning is $p \vee (q \rightarrow \neg r)$

then parentheses must be used.

Propositional Logic Summary

The Language of Propositions

- Connectives: Negation $\neg$ Conjunction $\wedge$ Disjunction $\vee$
Implication $\rightarrow$ Biconditional $\leftrightarrow$ Exclusive Or $\oplus$
- Truth Values
- Truth Tables

Logical Equivalences

- Equivalent Propositions

Exercise

13. Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

a) It is below freezing and snowing.

b) It is below freezing but not snowing.

18. Determine whether these biconditionals are true or false.

a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.

Exercise

33. Construct a truth table for each of these compound propositions.

a) $p \wedge \neg p$

c) $(p \vee \neg q) \rightarrow q$

Applications of Propositional Logic

Section 1.2

1.2.2 Translating English Sentences (optional)

Steps to convert an English sentence to a statement in propositional logic

- Identify atomic propositions and represent using propositional variables.
- Determine appropriate logical connectives

“If I go to Harry’s or to the country, I will not go shopping.”

- p : I go to Harry’s
- q : I go to the country.
- r : I will go shopping.

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example (opt)

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus **only if** you are a computer science major or you are not a freshman.”

One Solution: Let a , c , and f represent respectively “You can access the internet from campus,” “You are a computer science major,” and “You are a freshman.”

$(p \rightarrow q \quad p \text{ only if } q \text{ (仅当)})$

$$a \rightarrow (c \vee \neg f)$$

1.2.3 System Specifications(opt)

System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply cannot be sent **when** the file system is full”

Solution: One possible solution: Let p denote “The automated reply can be sent” and q denote “The file system is full.”

$$(p \rightarrow q \quad q \text{ when } p (\text{当}))$$

$$q \rightarrow \neg p$$

Propositional Equivalences

Section 1.3

1.3.1 Tautologies, Contradictions, and Contingencies

A *tautology*(恒真式) is a proposition which is always true.

- Example: $p \vee \neg p$

A *contradiction*(矛盾式) is a proposition which is always false.

- Example: $p \wedge \neg p$

A *contingency*(适然式) is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

1.3.2 Logically Equivalent(逻辑等价)

Two compound propositions p and q are **logically equivalent** if $p \leftrightarrow q$ is a tautology.

We write this as $p \leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.

Two compound propositions p and q are **equivalent** if and only if the columns in a truth table giving their truth values agree.

This truth table shows that $\neg p \vee q$ is equivalent to $p \rightarrow q$ (*conditional-disjunction equivalence*).

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$



Augustus De Morgan
1806-1871

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences₁

Identity Laws: $p \wedge T \equiv p, \quad p \vee F \equiv p$
(恒等律)

Domination Laws: $p \vee T \equiv T, \quad p \wedge F \equiv F$
(支配律)

Idempotent laws: $p \vee p \equiv p, \quad p \wedge p \equiv p$
(幂等律)

Double Negation Law: $\neg(\neg p) \equiv p$

Negation Laws: $p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$
(否定律)

Key Logical Equivalences₂

Commutative Laws: $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$

(交换律)

Associative Laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

(结合律)

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

Distributive Laws: $(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$

(分配律)

$$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$$

Absorption Laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

(吸收律)

More Logical Equivalences

TABLE 7 Logical Equivalences
Involving **Conditional Statements**.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences
Involving **Biconditional Statements**.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

1.3.4 Constructing New Logical Equivalences (opt)

We can show that **two expressions are logically equivalent by developing a series of logically equivalent statements.**

To prove that $A \equiv B$, we produce a series of equivalences beginning with A and ending with B.

$$\begin{array}{c} A \equiv A_1 \\ \vdots \\ A_n \equiv B \end{array}$$

Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs₁

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
$\equiv \neg p \wedge (p \vee \neg q)$	by the double negation law
$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
$\equiv F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
$\equiv (\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
$\equiv (\neg p \wedge \neg q)$	By the identity law for F

Equivalence Proofs₂

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology.

Solution:

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$	by truth table for $\rightarrow$
$\equiv (\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
$\equiv (\neg p \vee p) \vee (\neg q \vee q)$	by associative laws and commutative laws for disjunction
$\equiv T \vee T$	by truth tables
$\equiv T$	by the domination law

Propositional Equivalences Summary

1, Tautologies 恒真式, Contradictions 矛盾式, and Contingencies 适然式

2, Logically Equivalent

Two compound propositions p and q are **logically equivalent** if $p \leftrightarrow q$ is **a tautology**.

Two compound propositions p and q are **equivalent** if and only if the columns in a **truth table** giving their truth values agree.

We can show that **two expressions are logically equivalent by developing a series of logically equivalent statements**.

conditional-disjunction equivalence: $\neg p \vee q \equiv p \rightarrow q$

De Morgan's Laws $\neg(p \wedge q) \equiv \neg p \vee \neg q$

$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Exercise

4. Use truth tables to verify the associative laws

a) $(p \vee q) \vee r \equiv p \vee (q \vee r).$

9. For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.

a) $p \rightarrow \neg q$

Homework

Page 15: 19a, 34b, 35a

Page: 38: 11a, 20

19. Determine whether each of these conditional statements is true or false.

b) If $1 + 1 = 3$, then $2 + 2 = 4$.

c) If $1 + 1 = 3$, then $2 + 2 = 5$.

34. Construct a truth table for each of these compound propositions.

c) $p \oplus (p \vee q)$

d) $(p \wedge q) \rightarrow (p \vee q)$

11. Show that each of these conditional statements is a tautology by using truth tables.

b) $p \rightarrow (p \vee q)$

Show the conditional statement is a tautology by applying a chain of logical identities.